Behind the Wheel

A Data Story of Traffic Collisions in California

Introduction

Researching and understanding traffic collisions is critical for improving road safety, managing public policy, and reducing preventable injuries and deaths.

Main Research Question

What patterns and trends can be identified in California traffic collisions from 2020 to 2025, and how do various contributing circumstances affect the severity of crashes?

Hypotheses

1. Urban areas experience a higher frequency of collisions compared to rural areas due to higher traffic density and complex road networks.

2. Severe weather conditions such as rain, snow, fog increase the likelihood of vehicle collisions compared to clear weather conditions.

3. Collisions involving impaired driving and speeding result in a higher proportion of severe injuries.

Dataset

California traffic collision data from 2020 to 2025

- From the **California Open Data Portal**
- Reported through California Highway Patrol and the California Crash Reporting System (CCRS)

Total rows: Approximately 2.46 million rows per file

Data Cleaning: Standardized column names, merged files and years, map out corresponding county names to codes, cleaned unnecessary symbols

Limitations: Not all collisions were recorded in the dataset. Last report date November 2025.

Variables in the dataset

1) **Demographics**

Collision ID - Record of Crash
County - The county of California involving the collision
Gender & Age

2) **Context**

Weather Conditions - At time of crash
Temporal Information: Time of the day
Driver Behaviors: Human factors contributing to the collisions

3) **Consequences**

Number Injured (Count)
Number Killed (Count)
Severity Indicators - Impact of people who were injured/killed

Severity Score

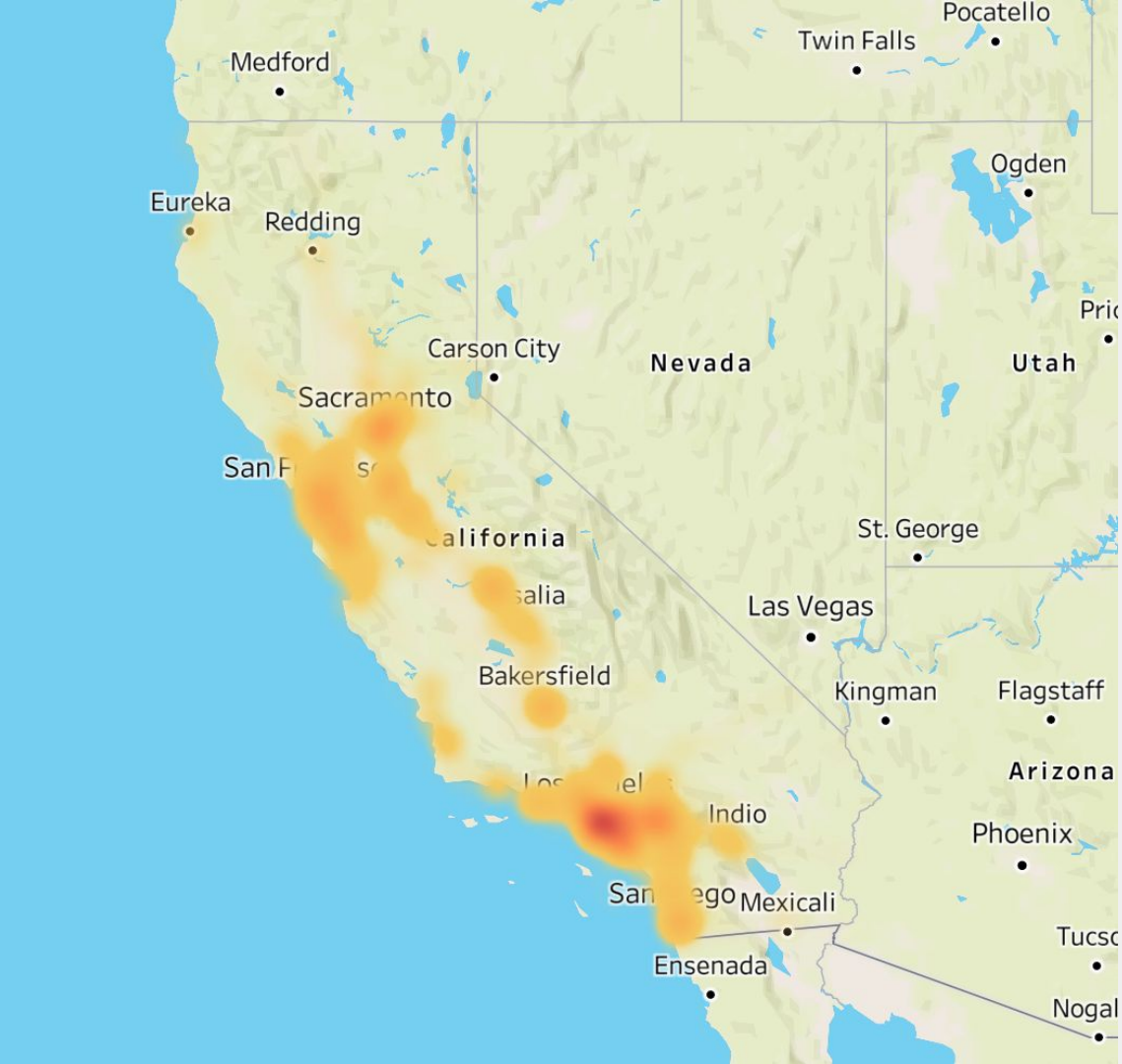
```
df["Severity"] = df["Number Injured"] + (3 * df["Number Killed"])
```

This is a common method used when looking at traffic safety analytics

Each injury = **1 point**
Each fatality = **3 points**

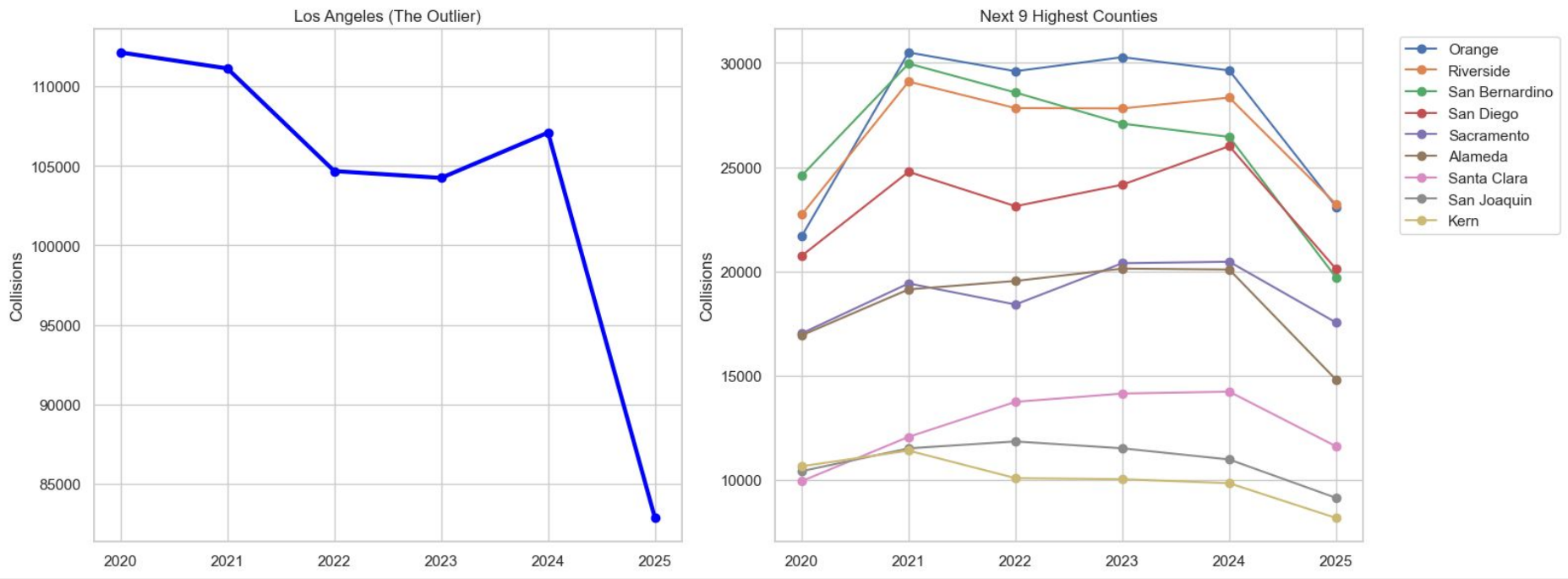
Geographic Distribution

Which regions or counties in California experience *the highest number of traffic collisions*, and *how does crash severity differ across locations?*

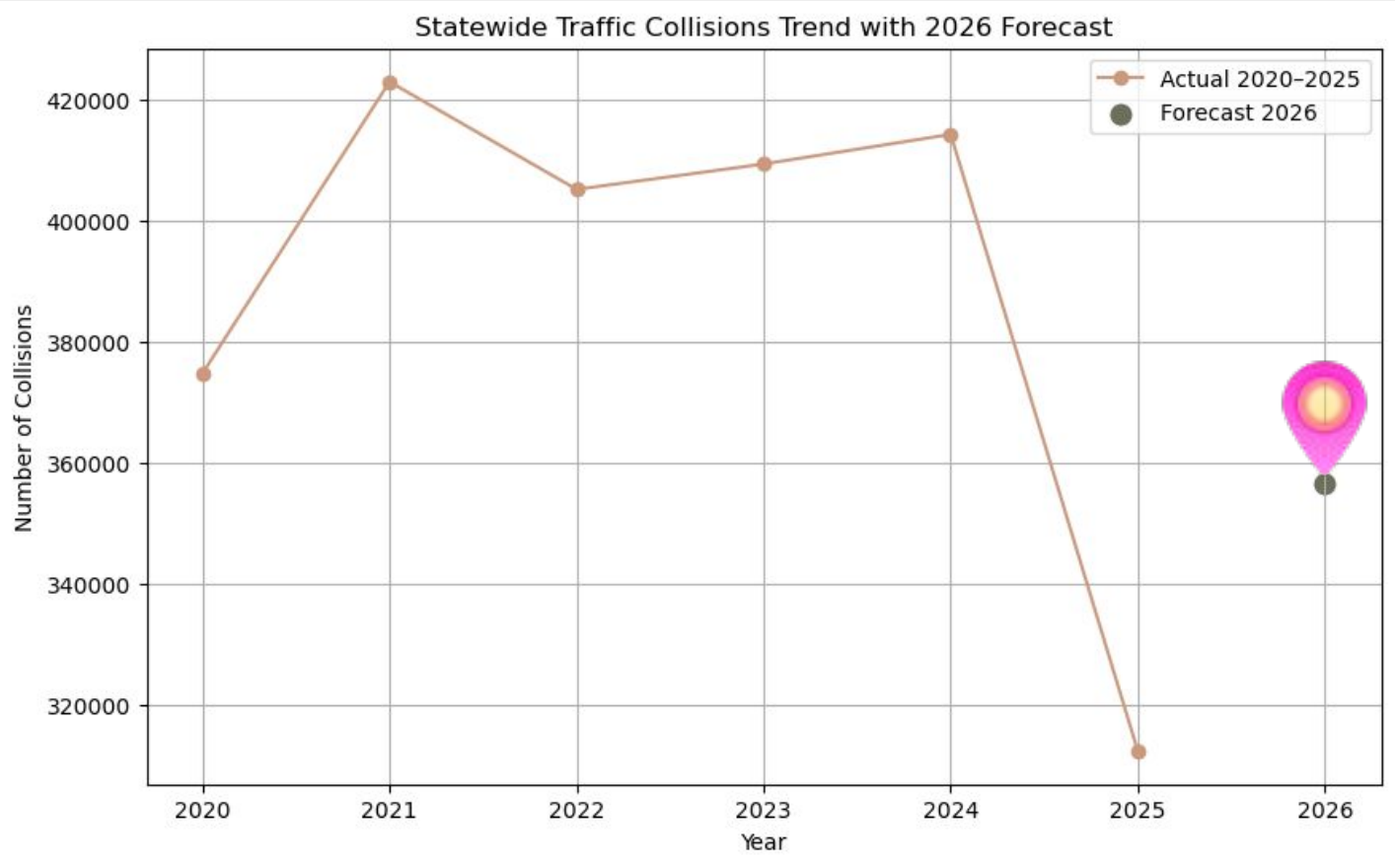


Density map of crashes in California

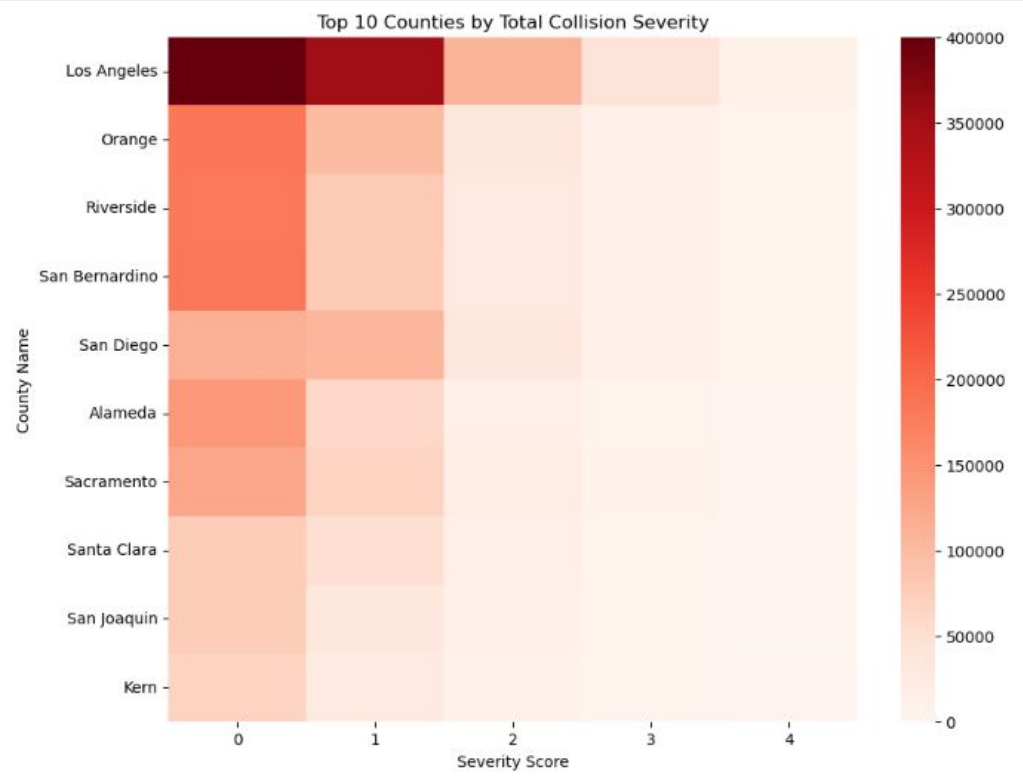
Top 10 Counties with Most Collisions



2026 Forecast Prediction



Counties with Highest Collision Severity

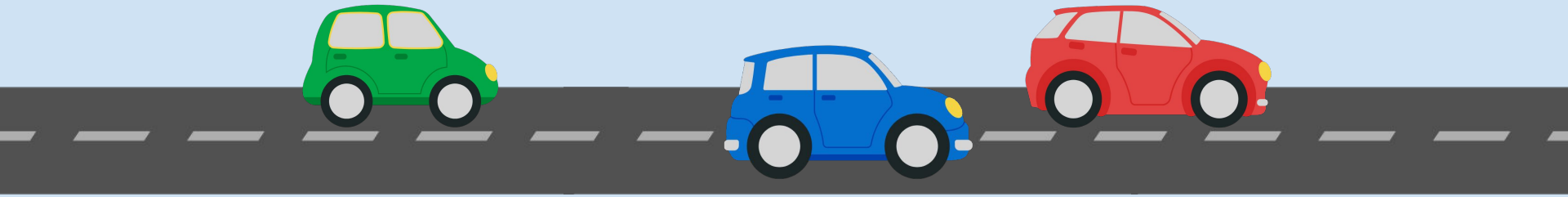


Implications:

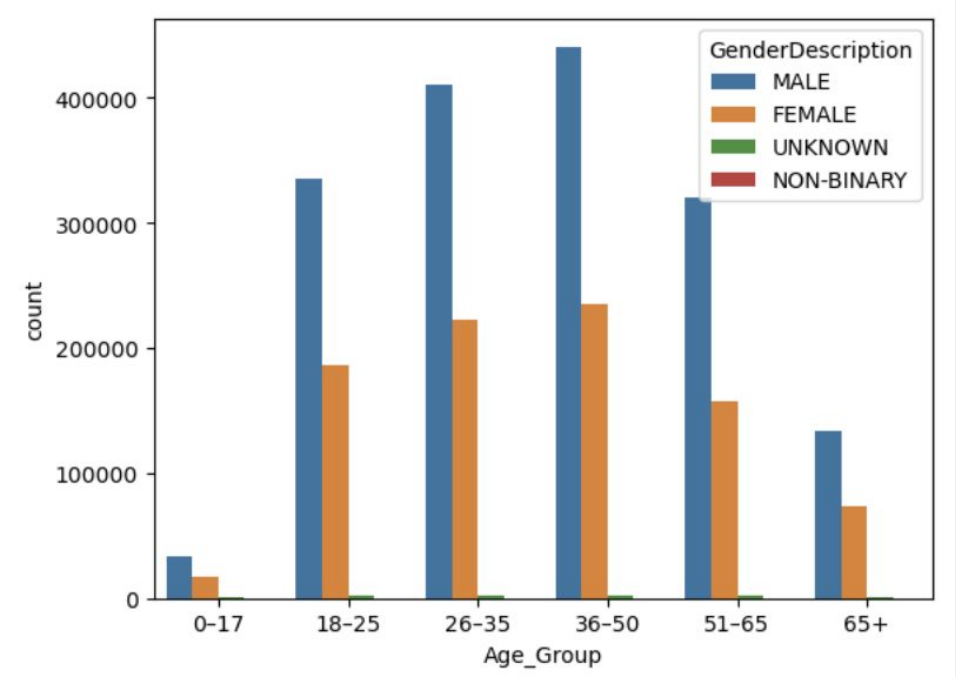
- Outlier
- Urban areas have the highest collision severity and count
- Mostly low-severity (0-1)
- Southern vs Northern California

Demographics Distribution

How does collision involvement vary by **age group** and **gender** from 2020–2025, and which demographic groups contribute the most to overall collisions?



Collisions by Age and Gender

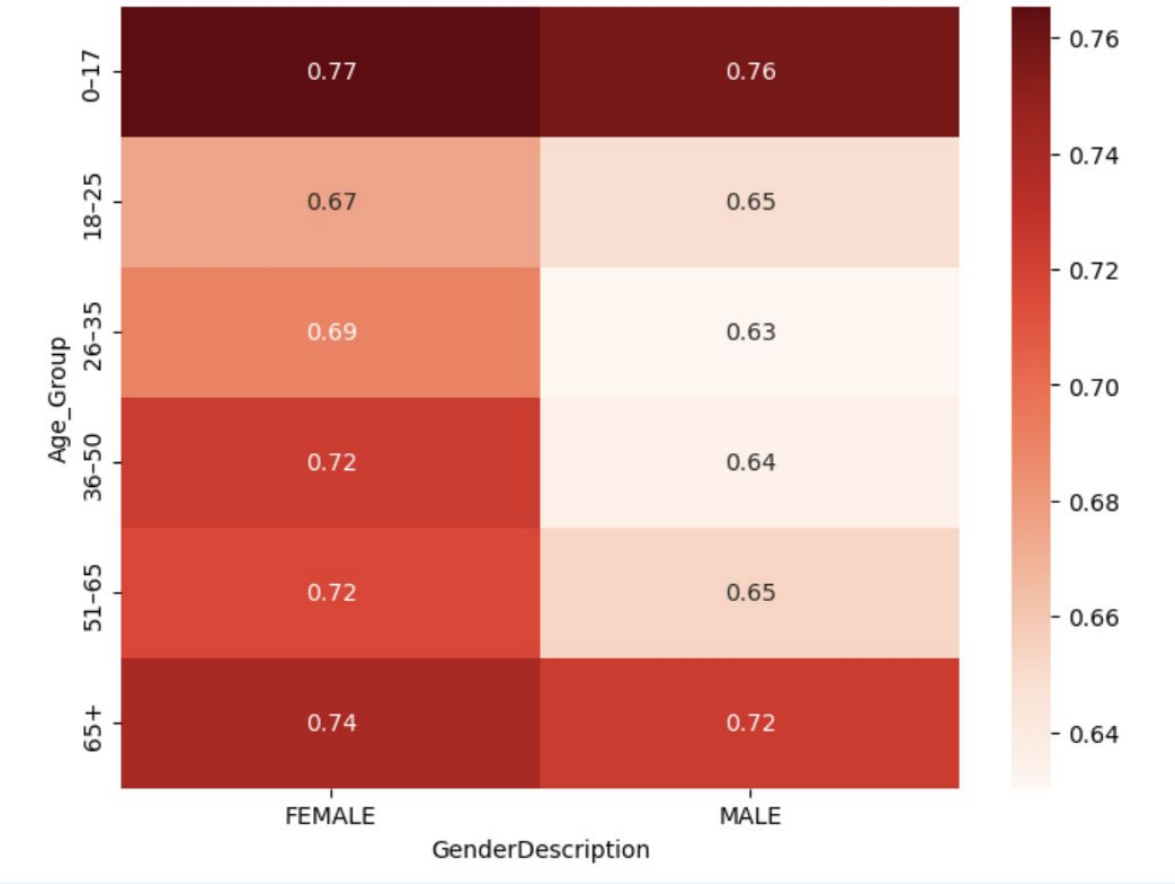


Limitations:

- Lack demographic totals
- Cannot calculate collision percentages by gender/age

Further research:
The ratio of male to female drivers is ~ 51% : 49%

Average Severity by Age Group 2020-2025



Implications:

Young drivers:

- Lack of experience
- More risk-taking
- Less skills

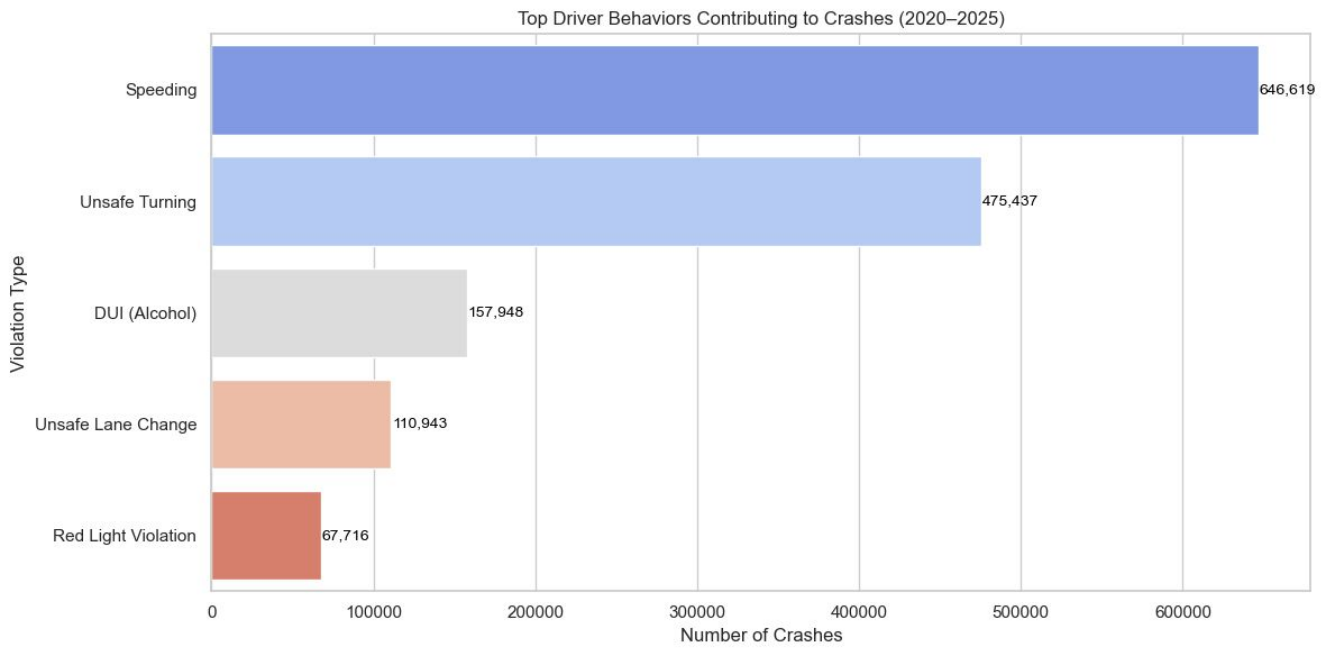
Elderly drivers:

- Physical limitations (vision, hearing, cognitive function, and mobility)
- Slower reaction times

Driver Behaviors & Collision Factors

What are the primary collision factors associated with crashes?

Which violations cause the highest accidents?



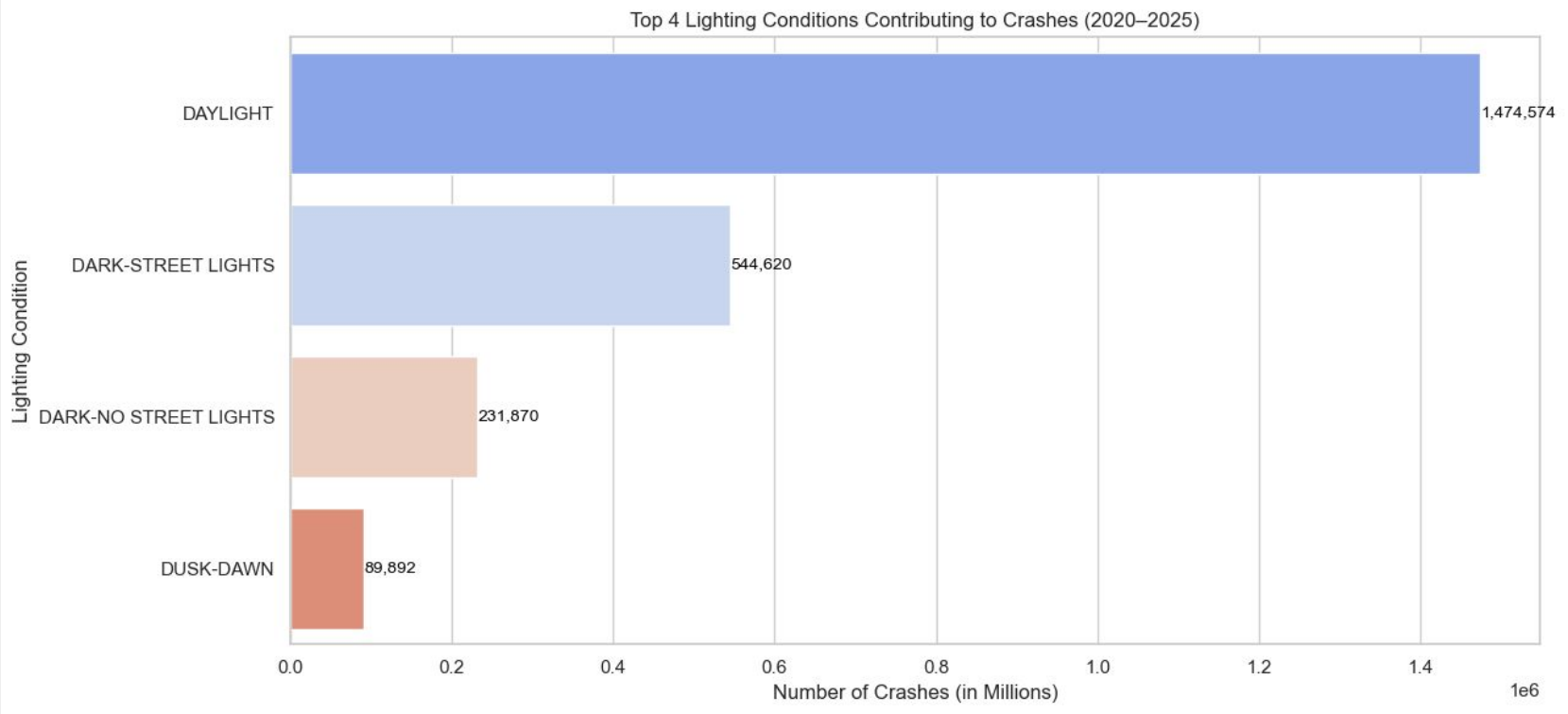
Limitations:

- PCF may be assigned differently by various officers
- There is only 1 primary factor is assigned per crash

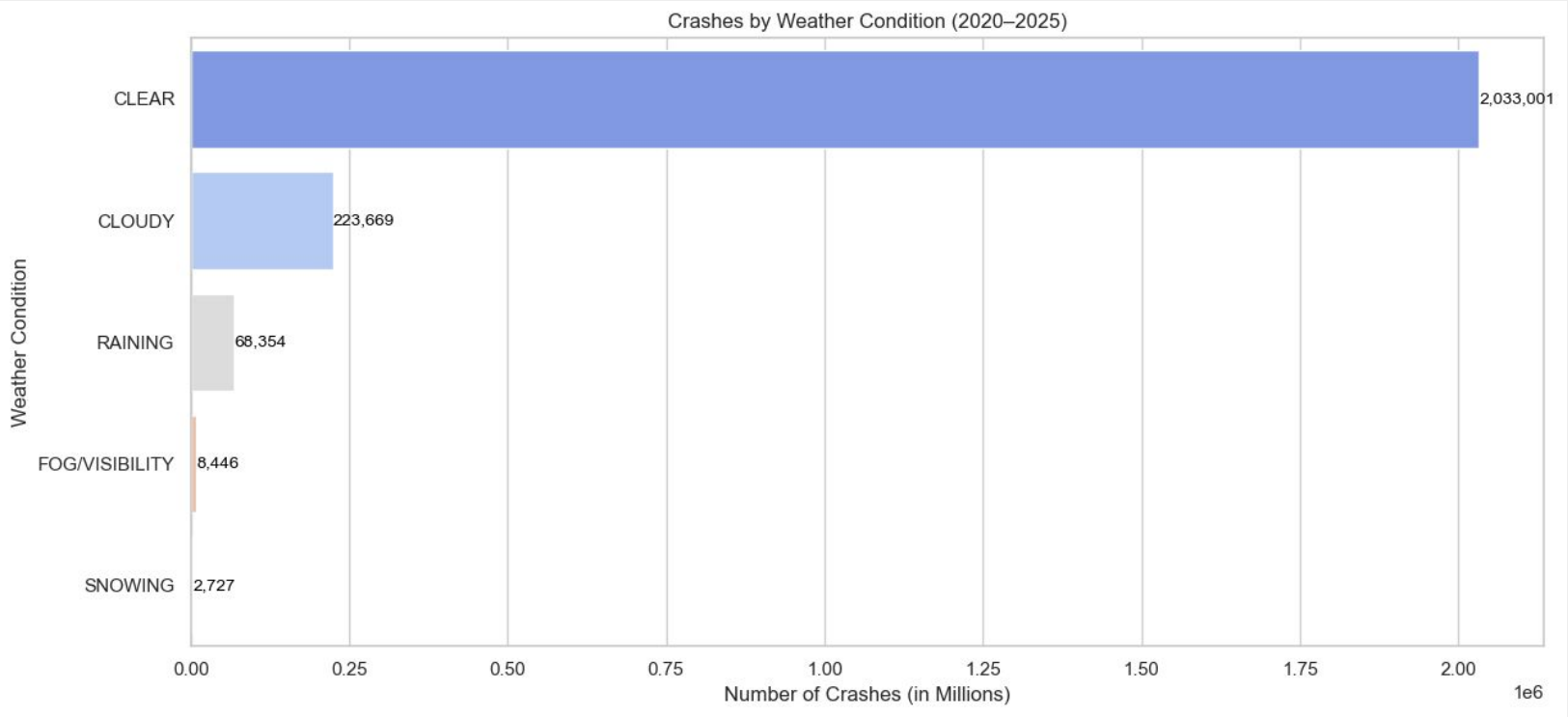
Collision Factors vs Severity Score

10	Driving on the wrong side of the road
8	Crossing double yellow lines
6.7	Driving Under the Influence (Alcohol)
6.5	Driving Under the Influence (Drugs)
6	Failure to yield at a stop

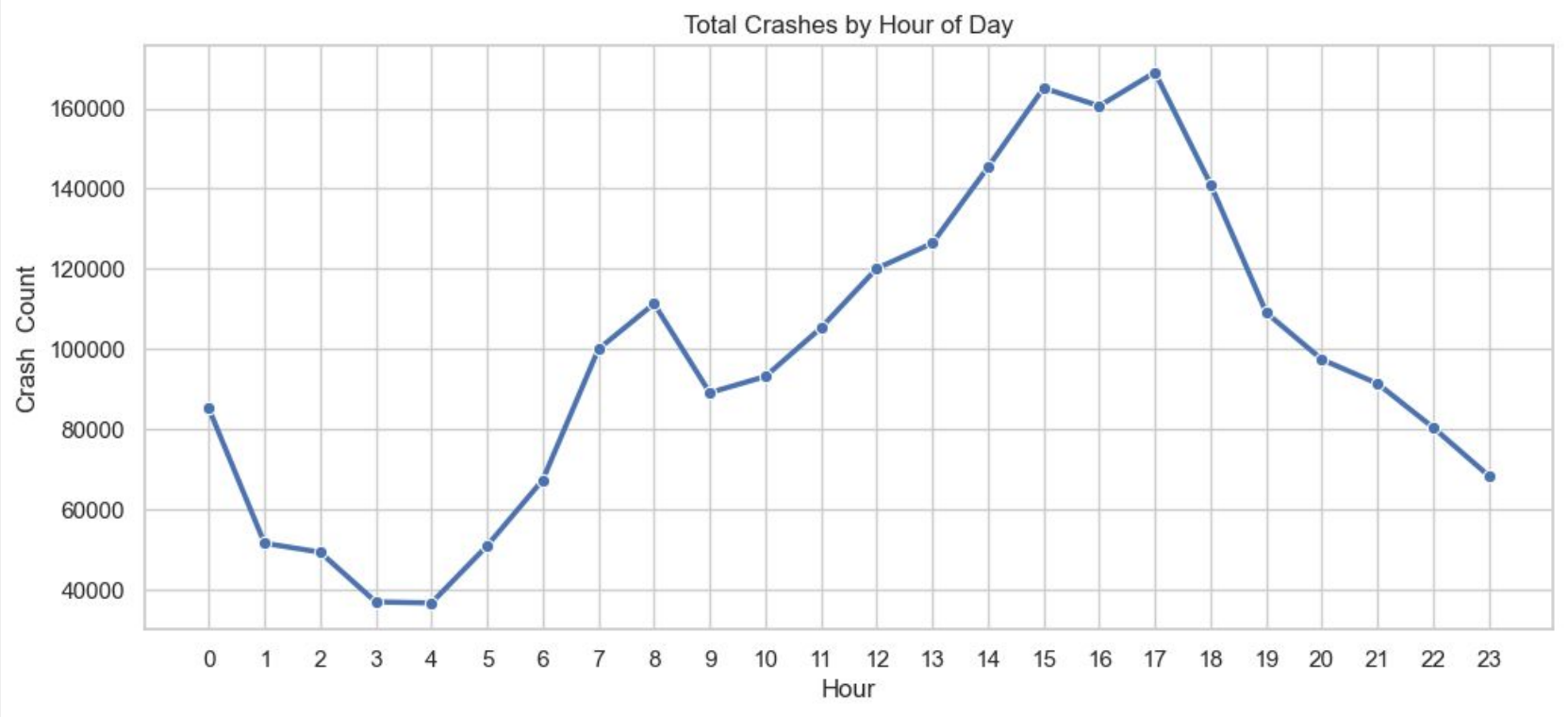
Do lighting conditions affect accidents?



Does poor weather affect crashes?



Total Crashes by Hour of Day



Conclusion 🚗

Actionable recommendations

- Add cameras in speeding hotspots
- Add physical lane separators and speed bumps
- Improve pedestrian crossings
- Increase lighting for lane visibility prioritize maintenance
- Investing in infrastructure and public transportation can help reduce overall crash risk